SoulSkipper: A Voice-Controlled Emotional Adaptive Game to Complement Therapy for Social Anxiety Disorder

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ABSTRACT

In this paper, we introduce SoulSkipper, a voice-controlled emotional adaptive game based on the use of Cognitive Behavioral Therapy (CBT) to complement therapy for social anxiety disorder (SAD). The game allows players to interact with non-player characters (NPCs) through voice input, which helps players identify negative thoughts and practice speech performance through 1) specific social scenarios, 2) emotional adaptation, and 3) real-time voice feedback. Through our initial evaluation, we find that the combination of game elements and voice interactions helps players improve their awareness of therapeutic skills associated with SAD and face social interactions in real-life scenarios more positively.

CCS CONCEPTS

• **Human-centered computing** → **Interaction design**; **HCI design and evaluation methods**.

KEYWORDS

social anxiety disorder; voice-controlled game; emotional adaptation; CBT

ACM Reference Format:
Xiaohan Li, Qixin Wang, Zishan Wang, Zeyu Jin, and Jia Jia. 2024. SoulSkipper: A Voice-Controlled Emotional Adaptive Game to Complement Therapy for Social Anxiety Disorder. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24), May 11–16, 2024, Honolulu, HI, USA.* ACM, New York, NY, USA, 7 pages. https://doi.org/10.1145/3613905.3650822

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CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
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ACM ISBN 979-8-4007-0331-7/24/05.
https://doi.org/10.1145/3613905.3650822

1 INTRODUCTION

Social Anxiety Disorder (SAD) is a common and widespread anxiety disorder. Patients typically experience fear and anxiety in one or more specific social scenarios and exhibit physical symptoms such as poor speech and increased heart rate [62]. This state of fear in social interactions can have multiple adverse consequences for patients, leading to disruption of social relationships, loss of confidence in life, and increased risk of suicide [14].

Cognitive Behavioral Therapy (CBT) is the main evidence-based treatment for SAD [18, 40, 50], in which psychotherapists face-to-face contact with patients to help them identify negative thoughts and face their fears by exposing them to specific social scenarios [45]. However, psychotherapy for SAD faces multiple barriers, including a lack of medical resources and privacy concerns, which make psychotherapy less accessible in clinical settings. In addition, the fear of social stigmatization among people with SAD makes it more difficult for them to seek professional therapeutic help [2, 64].

In this context, researchers have explored several digital interventions to improve SAD, thus providing anonymity and accessibility to therapy [6, 41]. Findings suggest that a range of digital interventions was not different from traditional face-to-face psychotherapy in terms of effectiveness [11, 21], however, they have shown their poor engagement and high attrition rates overtime [25]. Research has shown that engagement is associated with improved outcomes for digital interventions [59]. Therefore, there is a need for further research into the user experience and engagement of digital interventions to implement them more effectively among people with SAD, thus filling a gap that is not present in traditional therapies [25].

Serious games and gamification are considered as an attractive solution to increase user engagement [39]. Games and especially fun interactions may be more engaging than non-game digital interventions [25]. However, many current serious games for mental health are not evidence-based and have limitations in their design approach [31]. Moreover, research suggests that existing digital mental health interventions generally neglect human-computer interaction which negatively affects user engagement [5].

To further address these challenges, in this paper, we present a role-playing game based on voice-controlled and emotional adaptation to complement therapy for SAD. We adopt an evidence-based design approach using CBT to help SAD patients challenge negative thoughts and improve behaviors. Considering the speech symptoms of SAD, the game allows players to interact with non-player characters (NPCs) through voice input. It enhances user engagement by increasing the level of personalization of the game through 1) specific social scenarios, 2) emotional adaptation, and 3) real-time voice feedback.

2 GAME GOALS

Our game goals are as follows: **1)** Propose an innovative digital game to increase engagement with SAD patients to complement therapy. **2)** Support self-help CBT training through specific social scenarios that create a game place for players to engage in close to real-life situations, helping them to identify negative thoughts and learn strategies for coming up with positive thoughts. **3)** Combine emotional adaptation with timely feedback to help players reflect and adjust, and face social interactions in real-life scenarios more positively.

3 GAME DESIGN

We refer to the player/game/therapy model (P/G/T model) [39] for game design, this framework describes the game design as a two-step process. We began by independently analyzing all aspects of the game, including understanding the SAD players we were targeting, selecting the therapeutic elements to be embedded in the gameplay process, and defining the underlying genre of the game. We then combined the understanding of the player/game/therapy to generate our game.

3.1 Player Research

Our target players are adults with mild social anxiety disorder. First, we want to answer a research question: What are the anxiety conditions experienced by these SAD patients? To answer this question, we first collected and analyzed posts in online forums discussing social anxiety. Secondly, we organized a semi-structured interview with SAD patients (N=10).

We collected data from the Chinese online forum Douban[1], which allows members to create rich online groups for discussion. We obtained 15 online groups based on keywords such as "social" and "social phobia", and further filtered them by keywords such as "social anxiety", "SAD", and "fear" within the groups. To ensure timeliness, we screened to get a total of 4015 posts and comments with keywords between 2022-2023. 232 valid posts and comments were obtained.

To further identify user needs, we recruited 10 participants (3 males, 7 females, aged 23-29) from offline and conducted semi-structured interviews with them. After obtaining consent from the participants, we assessed the participants using the Liebowitz Social Anxiety Scale (LSAS) [42], and all of them demonstrated mild (>30) and higher social anxiety scores (M=63.2, scored 35-93). The interviews focused on their experiences of social anxiety, its resolution, and their expectations for assisted relief. Each interview lasted an average of 45-60 minutes.

The qualitative data were analyzed using an open coding approach [35], and two co-authors worked together to identify the initial themes. After coding and analyzing the data, we first derived 8 different social anxiety themes, such as "talking to strangers", "public speaking", and "talking in a group". We used keywords in different posts to identify similar themes. For example, multiple posts mentioning "meeting strangers", "meeting people for the first time", "when I'm in a crowd of strangers", etc., which were categorized under the "talking to strangers" theme. Secondly, we analyzed the anxiety reduction methods that SAD patients considered effective by repeatedly studying the supportive comments in online forums and the data from semi-structured interviews, we summarised them as "cognitive restructuring", "exposure therapy", "social skills training" etc. Finally, in the semi-structured interviews, participants expressed their expectations for assistive relief methods, including gamification ("The game tasks are less stressful than talking to a counselor") and timely feedback ("I need some objective feedback that tells me how I'm doing", "I'd like to know how effective I am at practicing my expression"). At the same time, we asked participants to rate their game-playing level using a Likert scale of 1-5 (1=low-high=5), and most of them expressed confidence(M=3.8, SD=0.75). These results helped us to define our game design.

3.2 Therapy Approaches

CBT uses cognitive techniques, as well as behavioral techniques, such as exposure therapy [53], to help people with SAD identify and challenge their anxious thoughts and change problematic behaviors. Research has shown that combining cognitive restructuring and exposure therapy is effective for SAD [7].

In our interviews, we asked participants to assess their knowledge of CBT (1=Low-High=5), most of them did not know about CBT (M=2, SD=0.89), and a small number of participants (N=2) would use methods similar to "thought records" [3, 56] to help them cognitively reorganize themselves ("I would reflect by keeping a journal to check my negative emotions "), but did not identify the concept of cognitive distortions [9]. Meanwhile, most users (N=8) emphasized the importance of exposure therapy ("To overcome social anxiety in addition to psychological counseling, it is more important to deliberately practice speaking, exercise social skills"). These findings are consistent with the research, therefore, we embedded these two CBT techniques in our game:

1) Cognitive Restructuring: CBT believes that many negative emotions arise from irrational thoughts [4]. In Cognitive Restructuring, the therapist helps patients: 1) understand his or her emotional state; 2) identify negative thoughts associated with the emotional state [16]; and 3) challenge these negative thoughts [27]. In our design, players will express emotions, identify negative thoughts, and make suggestions to non-player characters (NPCs) through an interactive form of dialog with NPCs. Players' verbal emotions will affect the NPCs' emotional state, thus helping players understand that using more positive thoughts to replace negative thoughts will reduce negative emotions. At the same time, players will also be guided by knowledge of concepts such as cognitive distortion in recognizing negative thoughts.

[1]Douban: https://www.douban.com/

2) Exposure Therapy: Exposure therapy is a behavioral technique in which patients face their fears under the guidance of a therapist [8]. Exposure therapy for social anxiety requires: 1) selecting specific exposure scenarios tailored to social interactions [55] and 2) effectively modeling the social scenarios [24, 60]. In our design, players will enter the specific social scenarios we set after completing the cognitive reconstruction task, engage in social tasks with NPCs in the form of a first-person perspective, and receive timely feedback, which will help players reflect and adjust their speech expression.

3.3 Game Overview

Our game, SoulSkipper, is a CBT-based role-playing game that combines voice control and emotion recognition. In this game, players will take on the role of "Soul Skipper" to help travelers who are lost in the "Sea of Emotions" to solve their emotional distress. Players will take a third-person perspective to understand the traveler's social anxiety situation, judge and identify the traveler's negative thoughts, and advise on voice input. Then, players will help the traveler reverse their emotional state through verbal emotion and will gain the "Rudder of Time" through the traveler's positive emotional changes, helping them return to the time before the emotional distress occurred. Finally, players will help the traveler perform social tasks from a first-person perspective to see the impact of positive behavior.

3.3.1 Game Genre. We chose "role-playing games" as our game genre. Role-playing is a commonly used tool in the treatment of SAD and can be used as an imaginative exposure to certain social situations [28]. At the same time, role-playing can also be better embedded with player-responsive narrative and emotional elements, increasing player empathy and engagement [1]. In terms of interaction, our game allows players to use voice input to talk to NPCs. This approach mimics the talk-back scenarios of actual therapy [61].

3.3.2 Game Play. Following the guidance of CBT, our game tasks will progress gradually according to the sequence of emotions, thoughts, and behaviors (Fig.1). Moreover, we embedded a fantastical narrative in the game to keep players continuously immersed in the gameplay [47]:

1) Accept Mission: Players navigate the "Sea of Emotions" and are guided by the "Nautical Map" to the location where the traveler is lost and accepts a new mission.

2) Enter Situation: Players will learn about the traveler's social difficulties and emotional distress through the traveler's narration and the display of images. Players will be guided by the "Navigation Light" to make judgments about irrational thoughts. If players' choices reflect any cognitive distortions, the "Navigation Light" will show players the concept of cognitive distortions and reasonable alternative thoughts.

3) Give Advice: Players will give advice to the traveler after identifying the traveler's negative thoughts. Players' verbal emotions will be detected. When players express positive emotions that shift the traveler's emotional state to positive, the traveler's emotions will provide the energy to help them return to the point before the emotional distress occurred.

4) Take action: Players use the "Rudder of Time" to go back in time, enter the traveler's perspective, and complete social tasks. Players will receive objective feedback on the quality of their speech expression.

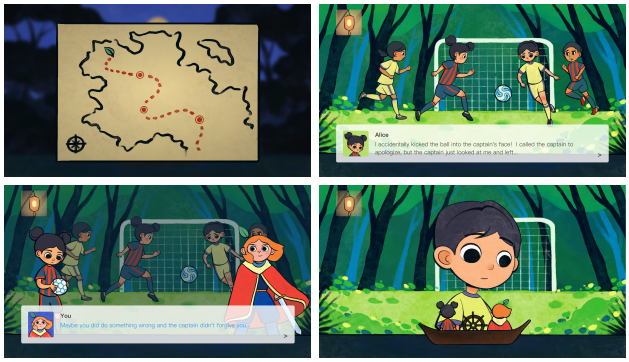

Figure 1: The main interface of the game

3.4 Design Process

Drawing on our research into game players, therapy approaches, and gameplay, we further integrated three aspects: storytelling, interaction design, and visual design.

3.4.1 Storylines. **1) Scenario Development:** The locus of SAD is often personalized [55]. Therefore, in our design, we focused on developing personalized social scenarios. Our story scenarios were all derived from the user research we completed in Phase 1. We ultimately identified 8 social scenarios including "talking to strangers" [36], "public speaking" [8], "talking in a group" [36], "talking in a job interview", "talking to people in authority" [32], "ordering a meal", "shopping at the mall", and "speaking at a conference". These scenarios were based on the themes we identified in the first phase and were further reviewed to generate stories.

2) Story Generation: After identifying the social scenario in which the stories took place, we further coded the themes in each scenario. For example, the scenario "talking to strangers" contains several sub-themes such as "I was afraid to speak", "I was afraid to look at each other", "My facial expressions and movements started to be unnatural", etc. These rich narrative details provide materials for our story generation. Based on these materials, we adapted our stories in conjunction with the Freytag's Pyramid [33] (a theatre structure in a five-act framework) and made sure they did not point to any particular original story.

3) Narrative Structure: Each story we generate will be treated as an independent chapter, and each chapter follows a consistent structure, using the ABCDEF model of CBT [48] as a framework. Referring to this framework, we divided the episodes into the following six sections: social anxiety situations (Activating Event), negative thoughts (Beliefs), behavioral and emotional consequences of negative thoughts (Consequences), counterarguments to negative thoughts (Disputation), shifts in positive thinking patterns (Effective New Beliefs), and behaviors (Future Action). Fig.2 shows the storylines for the chapter "Talking to Strangers".

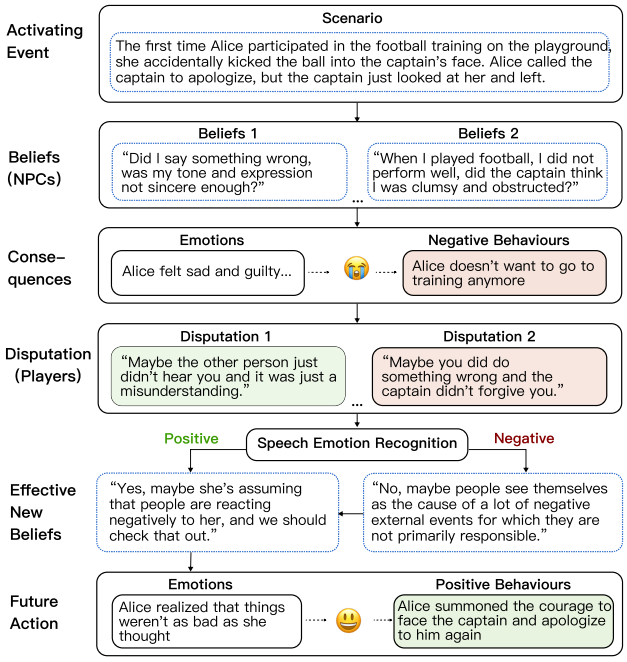

Figure 2: Mapping the storylines to the ABCDEF-model

3.4.2 Interaction Design. **1) Emotional Adaptation:** The emotions that players experience during gameplay can stimulate their immersion and engagement [13, 65]. In our design, we chose to embed an element of emotional adaptation to create a personalized gaming experience for the player [12], as well as to help players understand the principles of CBT and how emotions affect our thoughts and behaviors.

In previous studies, various modalities have been used to detect the player's emotional state, including physiological signals [26], voice intonation [19], and facial expressions [57]. In our design, we encourage players to communicate with NPCs through voice input, we use acoustic and textual features to recognize players' emotional state, which will be detected as positive or negative.

We adapt to the player's emotional state in two ways (Fig.3): 1) Adaptation of NPCs' emotional appearance: NPCs will display corresponding positive or negative emotional states according to the player's emotion; 2) Adaptive automation of task content: the game will provide automated assistance and prompts according to the player's emotion to encourage the player to shift to a positive mindset. The automatic feedback is selected from a predetermined list. When the player's emotion is detected as negative, the game guide will provide positive suggestions. Through this design, we enhance the connection between the player and the game character, helping the player visualize how they should support and encourage their shy friend, thus helping them recognize their irrational mindset. At the same time, the adaptive nature of the content encourages players to advance the game's tasks in a more dynamic way [43].

2) Objective Feedback: SAD patients often exhibit behaviors such as poor speech and avoidance of eye contact [4, 10], these

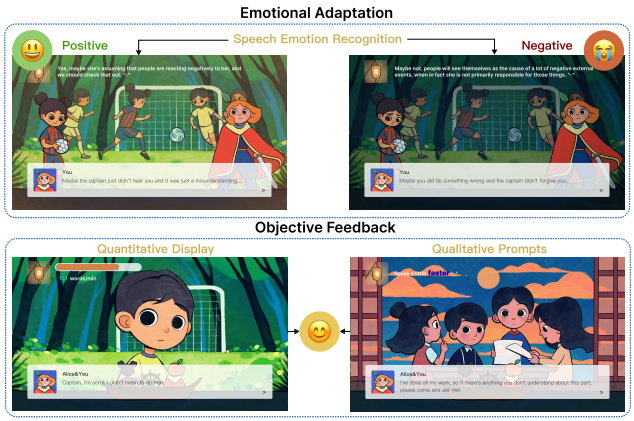

Figure 3: Emotional Adaptation and Objective Feedback

nonverbal behaviors can interfere with the delivery of information, and exacerbate the patient's negative emotions. For example, speaking too fast can limit the listener's understanding, and too low is not conducive to clarifying a message [29]. Exposure therapy helps patients gradually overcome fears and improve their behavior through planned exercises. In our design, we help players understand and adjust their vocal performance by extracting their volume and speech rate values and providing real-time personalized feedback.

In the game story, players should help the protagonist (the traveler) go back to the past, complete social tasks, and face the game's conversational objects from a first-person perspective (we set up one or more NPCs according to different social scenarios). We designed specific dialog quests for each chapter of the game and provided players with prompted texts that players could read or speak in his or her own way. When players speak, the game interface displays real-time objective feedback metrics, including volume and speech rate. When a speech task is completed, players will receive nonverbal feedback (a warm smile) from the NPCs, which encourages players to complete the task.

For the visualization of objective feedback, we considered two forms of display (Fig.3): 1) real-time quantitative display: dynamic volume bar and speech rate value; 2) qualitative indicator prompts: we provided alerts to players concerning the optimal volume (50-65 dB) and speech rate value (150-180 words/min) [37] for the Chinese language (which is the native language of all our players). When players' voice performance was not within these thresholds, we provided them with textual reminders of "volume is a bit low/high" and "speech speed is a bit slow/fast". Considering the amount of information players can receive during a conversation [44], we set an interval of 20s [58] between these text alerts. During the design process, we will use an iterative design methodology to invite players to experience each of the above two forms of objective feedback and improve our design through players' evaluation results.

3.4.3 Visual Design. We refer to previous research and use positive visualization strategies to help SAD players reduce the negative experiences caused by stress. We embedded natural elements such as plants and water in the game scene, research has shown that these

visual elements have a positive impact on the player's emotional state [22]. For the overall visual background, we also chose a visual scheme with bright colors and high saturation to reduce the player's associations related to negative emotions [34].

4 TECHNICAL SUPPORT

In this game, we encourage players to interact with the game through voice input. To achieve this goal, we record their speaking voice during gameplay through a microphone and extract audio and text features to recognize their emotional changes. At the same time, we provide players with objective feedback on speech quality by measuring the speed and volume of the input audio.

4.1 Speech Emotion Recognition

4.1.1 Methodology. Initially, we extract text content through the pre-trained model Whisper [51]. Then, multiple pre-trained embeddings are employed to abstract features for multi-model SER-Fuse [49], including BERT [23] (Bidirectional Encoder Representations from Transformers) for text embedding feature vector generation and x-vector [54], ECAPA [52], d-vector [63] for audio embedding feature vector generation. After obtaining multiple feature vectors, a linear combination of all features into a single feature vector is achieved through concatenation. Subsequently, this fused feature vector is fed into an FC (Fully Connected) layer to generate the final prediction, providing results for negative, neutral, and positive sentiment in three-class emotion prediction (Fig.4).

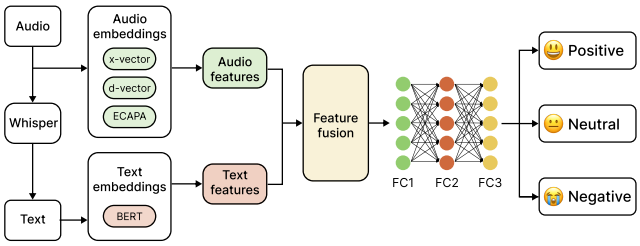

Figure 4: The workflow of our speech emotion recognition

4.1.2 Implementation. The model is trained for 200 epochs on the CH-SIMS v2.0 [38] dataset, after which we input the collected 226 real player data into the model to obtain prediction results. It can be observed that due to the generally calm tone of users in everyday speech, the model sometimes exhibits errors in distinguishing between negative and neutral emotions. However, the total accuracy is still 71.68%, demonstrating a reasonable level of emotion recognition.

4.2 Real-time Feedback on Speed and Volume.

The calculation of the speech speed is implemented using the Tencent Automatic Speech Recognition [30, 66]. Based on real data testing, we found that when the duration of the input audio reaches three seconds, stable and accurate speed results can be obtained. To provide more immediate feedback, we record the speech and calculate the speed at three-second intervals. Then, we report it to the player as a numerical value (words/minute). The player's volume is obtained through Audioworklet [20], which gets the microphone volume in real-time and maps the changes in the volume value through a bar graph.

5 PLAYER EVALUATION

We invited 10 participants who were recruited in user interviews to participate in the initial evaluation, each player was asked to complete 3 different game scenarios. In conjunction with our game goals, we initially evaluated our game prototype through the following three components: 1) assess whether our game was engaging for players through the System Usability Scale (SUS) [15], as well as the Engagement Scale adapted from the User Engagement Scale (UES) [46] and the Narrative Engagement Scale [17]; 2) assess whether our game was effective for player's CBT skills learning through the CBT Knowledge Test before and after the game; 3) assess whether the interactive features we designed were appropriate by inviting users to rate them (1=Low-High=5). After completing the questionnaire, we invited players to participate in a semi-structured interview to discuss their perceptions of our game.

Overall, our system had an average usability score of 80.75% and an average user engagement score of 3.73 (SD=0.43), which suggests that our system is attractive to players. Some players (N=3) believe that "the game gives some close-to-life scenarios, and through the experience of the game I can have better control of my emotions in actual social scenarios". The mean scores on our pre-game and post-game CBT knowledge tests (10 questions) were 4.0 (SD=1.53) and 7.17 (SD=1.5) respectively, which suggests that players improved their understanding of CBT skills through our game.

We designed two voice interactions: emotional adaptation and objective feedback. Initial evaluation results showed that players favored the emotional adaptation (M=4.08, SD=0.76), believing that "after I said what I thought, I felt that the character of the story got my help and it was very fulfilling". For both visual forms of objective feedback, users preferred real-time quantitative display (M=3.25, SD=1.64), and some players (N=5) felt that "the visual bars gave me a sense of participating in a positive challenge", but others (N=2) felt that this approach would "stress them out ". Therefore, continued discussion of the balance between volume, speed, and participant comfort is necessary. Players also suggested that they needed a reminder of the qualitative indicators (M=2.83, SD=1.07). However, the current textual reminder was "not very attention-grabbing"; therefore, we need to consider more engaging and gentle approaches in our future work.

6 FUTURE WORK

Our future work will focus on two main areas:

1) Randomized Controlled Trial: First, we plan to conduct an iterative design to optimize the user experience and consider cultural adaptation. Second, due to the limitations and potential biases associated with the small sample size of the current experiment, we plan to invite more users to conduct a randomized controlled trial, set up a control group, and randomize participants. In addition, CBT requires consistency, and we plan to study the long-term effects of using this game; therefore, we will ask users to complete the full range of game scenarios we have developed and conduct repeated measures in future experiments. These improvements aim to make

the game a more effective digital mental health tool for SAD treatment, offering a creative solution to current treatment challenges and potentially broadening its applicability and effectiveness in the mental health domain.

2) Improvement of Technical Methods: To further improve the accuracy of speech emotion recognition, we are considering additional pre-trained speech emotion models as auxiliary resources. By combining the results of these models with the SER-Fuse model in a weighted manner, our goal is to obtain more accurate comprehensive outcomes, focusing on distinguishing between negative and neutral emotions. Additionally, we recognize that the dataset's quality significantly influences the model's training quality. Therefore, we plan to augment the dataset with high-quality data to enhance the accuracy of emotion classification.

ACKNOWLEDGMENTS

We thank all participants for their participation.

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